

Literature Benchmark Brief: Pao_2011_CSM

Title: Control of wind turbines: Approaches, challenges, and new advancements

Authors: L. Y. Pao, K. E. Johnson (2011)

Venue: IEEE Control Systems Magazine | **DOI:** 10.1109/MCS.2010.939962

Cluster: Cluster C: Aerodynamics & Control Cliff

Core Theoretical Mechanism & Model Formulation

Survey of variable-speed pitch-regulated wind turbine control regimes and actuator limits

Empirical Scale & Experimental Dataset

Utility-scale variable-speed commercial turbines

Relevance to IEEE TSTE & Sustainable Energy Systems

Authoritative IEEE reference on operational control regions across wind speeds

Identified Gaps & Operational Vulnerabilities

Does not address grid pre-dispatch reserve sizing under imperfect telemetry

Role in Manuscript Defense & Reviewer #2 Proof

Establishes standard IEEE terminology for Region 2 (MPPT) and Region 3 (Pitch)